

CloudWatch AI Control Plane

Engineering Handoff | 4 System Components | 18 Acceptance Criteria | 4-Phase Rollout | 10 Risks | 6 Open Questions

Executive Summary

New top-level CloudWatch console page giving ops and compliance teams a single-pane view of all AI workloads across their AWS Organization. Addresses the gap: enterprises deploying AI across multiple accounts have no centralized way to monitor, govern, or audit. Proposed timeline: **12-week build to GA** (internal dogfood Week 4, beta Week 8).

1. Engineering Spec

System Components

#	Component	Type	Key Details
1.1	AI Workload Discovery Service	New	CloudTrail events -> EventBridge -> Lambda -> DynamoDB. Cross-account v
1.2	Governance Engine	New	Bedrock Guardrails API + SageMaker Model Cards. Rule-based maturity scor
1.3	Cost Aggregation Service	Extension	Cost Explorer API with AI service filters. Per-workload/model/account/team a
1.4	Console Frontend	New page	React + Cloudscape. URL: /cloudwatch/home#ai-control-plane. React Query

API Contracts

Endpoint	Method	Description
/ai-control-plane/workloads	GET	List all workloads with health, guardrail, cost
/ai-control-plane/workloads/{id}	GET	Workload detail: metrics, guardrails, cost, timeline
/ai-control-plane/workloads/{id}/timeline	GET	CloudTrail events for workload
/ai-control-plane/maturity	GET	Org maturity score, level, recommended actions
/ai-control-plane/maturity/config	PUT	Update maturity config (admin only)
/ai-control-plane/reports/compliance	POST	Async compliance report (returns S3 URL)
/ai-control-plane/settings	GET/PUT	Cross-account, discovery, notification prefs

Performance Requirements

Metric	Target	Rationale
Page load (100 workloads)	< 2s P95	CloudWatch console SLA
Split panel render	< 500ms P95	Perceived instant interaction
Property filter response	< 200ms P95	Client-side on cached data

Metric	Target	Rationale
Cross-account discovery	< 5 min	Near-real-time via EventBridge
API availability	99.95%	CloudWatch console target

2. Acceptance Criteria (18 total)

Fleet Overview

- GIVEN 14 workloads across 4 accounts WHEN navigate to AI Control Plane THEN see 4 summary cards within 2s
- GIVEN 3 new workloads since last visit WHEN page loads THEN Flashbar shows 'new workloads detected'
- GIVEN table displayed, no filters WHEN page loads THEN table sorted by health (error first)

Table Interactions

- GIVEN table WHEN click column header THEN sorts asc on first click, desc on second
- GIVEN Property Filter WHEN click input THEN suggestion dropdown with service/health/guardrail/account
- GIVEN filter 'Health=Error' applied WHEN filter active THEN only error workloads shown + token visible
- GIVEN 2 filters active WHEN click X on one THEN that filter removed, table updates

Split Panel Detail

- GIVEN table WHEN click any row THEN bottom split panel (40% height) opens with 4 tabs
- GIVEN split panel Overview tab WHEN view THEN see service, model, account, team, metrics, chart
- GIVEN workload health=error WHEN view Overview THEN inline error alert with explanation
- GIVEN workload guardrail=none WHEN view Guardrails tab THEN warning alert + 'Enable Guardrail' link
- GIVEN split panel open WHEN click X THEN panel closes, row deselected

Maturity & Edge States

- GIVEN maturity section WHEN view THEN see level (1-5), progress bar, 3 recommended actions
- GIVEN no workloads discovered WHEN page loads THEN empty state with 'Enable AI Control Plane' CTA
- GIVEN 2 accounts with permission failures WHEN page loads THEN error Alert + available accounts still render
- GIVEN page loading WHEN data fetching THEN skeleton loading for table, spinner for charts

Date Range & Export

- GIVEN Date Range Picker WHEN click '30d' THEN all page data updates to 30-day window
- GIVEN Actions dropdown WHEN click 'Generate Compliance Report' THEN async generation + Flashbar notification

3. Phased Rollout (12 weeks to GA)

Phase	Scope	Duration	Success Gate	Rollback
0 - Dogfood	APM team only, 2-3 accounts	2 wks	No P0/P1, <2s P95, >95% accuracy	Data integrity issue
1 - Beta	10-15 enterprises, opt-in	3 wks	<1% API errors, CSAT>4.0, handle 500+ orgs	CSAT < 3.5 or >40% disable
2 - Preview	All customers, Preview label	4 wks	Coverage delta +5pp in 50% orgs, WAU > 1000	Async WAU backslash or P0 security
3 - GA	All customers, all regions	Ongoing	All PRD metrics at target, <0.1% WAF	WAF rate

4. Success Metrics

Metric	Type	Target	Instrumentation
Guardrail Coverage Delta	North Star	+15pp in 90d	CW custom metric per org (15min)
Weekly Active Users	Engagement	5,000 by month 3	CloudWatch RUM page views
Time-to-Insight	Efficiency	<30s fleet health	RUM timing: load to split panel
Maturity Progression	Outcome	20% orgs +1 level in 90d	MaturityLevel metric (daily)
Cost Optimization Actions	Revenue	10% users/mo take action	Click tracking on CTAs

Phase Gates: v2 decision at Month 6 (if delta <+10pp AND WAU <2K, narrow to Bedrock-only). Pricing decision at Month 4 (if WAU >3K + retention impact, evaluate tiered pricing).

5. Meeting Deck Outline (27 min)

#	Section	Time	Speaker
1	Problem Recap (Maya persona, JTBD)	2 min	PM
2	Competitive Context (ServiceNow, Datadog, Dynatrace)	2 min	PM
3	Solution Overview (4 capabilities, scope boundary)	3 min	PM
4	Design Walkthrough (LIVE PROTOTYPE DEMO)	5 min	PM + UX
5	Engineering Spec (components, APIs, data model)	5 min	Eng Lead
6	Phased Rollout (4 phases, success gates)	3 min	PM + Eng
7	Open Questions (6 items for eng input)	5 min	All
8	Next Steps (Sprint 1 scope, ownership)	2 min	PM

6. Consolidated Risk Register

#	Source	Risk	L	I	Owner
R1	Gandalf	CloudTrail volume at scale (>1M events/day) causes >5min latency	M	H	Eng Lead
R2	Gandalf	SageMaker custom endpoint discovery accuracy <90%	M	M	PM
R3	Gandalf	Dashboard P95 >8s with >200 workloads	L	H	Eng Lead
R4	Gandalf	Free tier creates expectation; hard to price later	M	M	PM+Mktg
R5	PRD	ServiceNow launches cross-cloud before our GA	M	H	PM
R6	PRD	IAM cross-account setup friction (>50% abandon)	M	H	Eng Lead
R7	Design	Split panel content overload (4 tabs too many)	L	M	UX
R8	Design	Maturity score perceived as judgmental	L	M	PM+UX
R9	Technical	DynamoDB cost at scale (>10K workloads)	L	L	Eng Lead
R10	Technical	Bedrock Guardrails API rate limits	M	M	Eng Lead

7. Open Questions for Engineering

#	Question	Owner	Deadline
OQ1	Sustain <5min discovery at 1M+ CloudTrail events/day? (SageMaker EventBridge Pipes pattern)	Eng Lead	Sprint 0
OQ2	DynamoDB vs Timestream for workload registry?	Eng Lead	Sprint 0
OQ3	SageMaker endpoint discovery completeness via CloudTrail? Eng + SM team	Eng + SM team	Sprint 1
OQ4	Bedrock batch guardrail status API (feature request)?	PM	Week 3
OQ5	CloudFormation StackSet for cross-account IAM roles?	Eng + IAM	Sprint 1
OQ6	Regional availability for v1 (all vs subset)?	PM + Eng	Sprint 1

Appendix: Pipeline Artifacts

Stage	Artifact	Status
1 - Research	research-v1.md	Complete (6/6)
2 - PRD	prd-v1.md + gandalf-fixes	Complete (skill v0.2.0)
3 - Gandalf	gandalf-evaluation-v1.md	Complete (7/10, 3 resolved)
4 - Design	design-spec-v1.md	Complete (4.4/5 avg)
5 - Prototype	prototype-v1.html	Complete (18/18 checks)
6 - Launch	launch-readiness-v1.md	This document